PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE NAME											
CT GROUP	**1**	**6**				INDEX NUMBER					

CHEMISTRY **9729/03**

Paper 3 Free Response **15 September 2017**

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer all questions in this section.

1 Compounds containing halogens form a large class of chemicals, which are of industrial importance and are synthesised in huge quantities worldwide.

(a) Phosgene, $COCl_2$, is a gas which was produced during World War I, and was used as a chemical weapon. One method to produce phosgene from trichloromethane is as shown below.

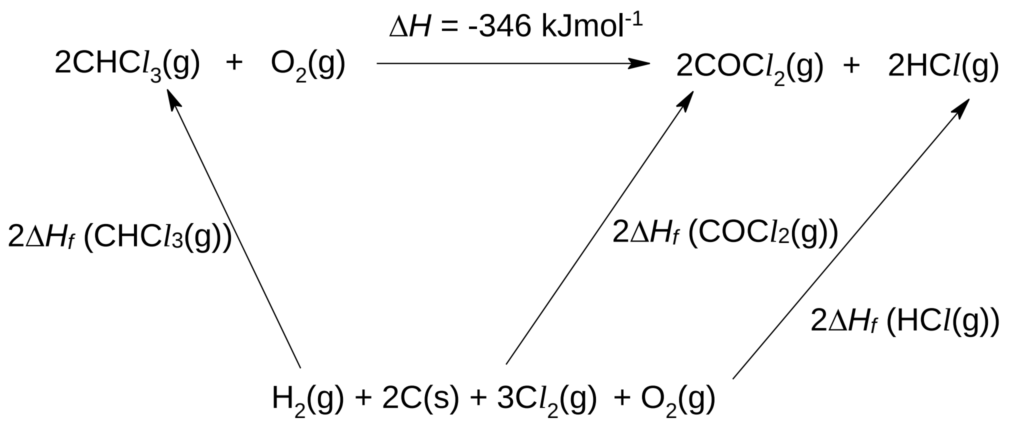
$$2CHCl_3(g) + O_2(g) \rightarrow 2COCl_2(g) + 2HCl(g) \qquad \Delta H = -346 \text{ kJ mol}^{-1}$$

Using the data below, calculate the standard enthalpy change of formation of $HCl(g)$.

standard enthalpy change of formation of $CHCl_3(g)$	-103 kJ mol^{-1}
standard enthalpy change of formation of $COCl_2(g)$	-184 kJ mol^{-1}

[1]

no need cycle



By Hess' Law,
$2\Delta H_f\,(CHCl_3(g)) + \Delta H = 2\Delta H_f\,(COCl_2(g)) + 2\Delta H_f\,(HCl(g))$
$2(-103) + (-346) = 2(-184) + 2\Delta H_f\,(HCl(g))$
$\Delta H_f\,(HCl(g))$= **$\underline{-92.0 \text{ kJ mol}^{-1}}$**

(b) Aqueous hydrochloric acid is another chemical with many industrial uses. It can be obtained by reacting hydrogen gas and chlorine gas in the presence of UV light, followed by dissolving the hydrogen chloride gas in water.

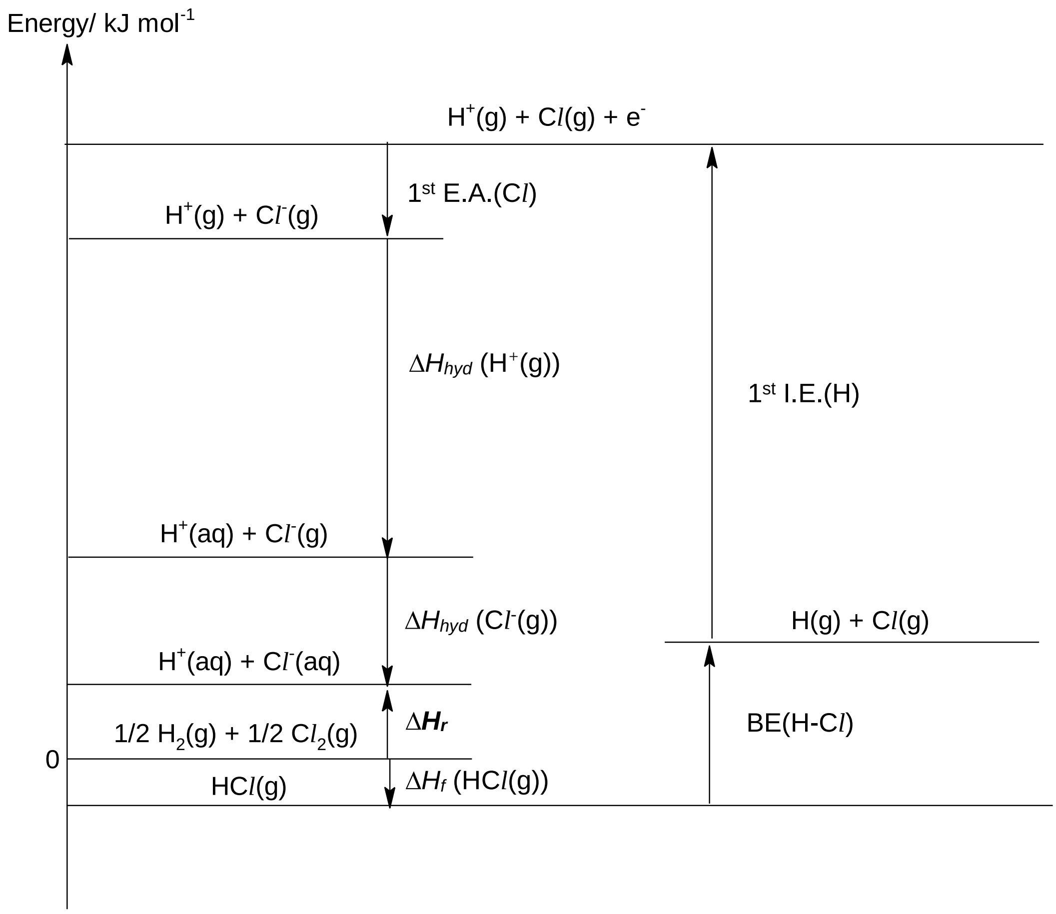
The equation for this reaction is as shown below.

$$\tfrac{1}{2}H_2(g) + \tfrac{1}{2}Cl_2(g) \rightarrow H^+(aq) + Cl^-(aq) \qquad \Delta H_r$$

Using the Data Booklet, the data in the table below, and your answer in **(a)**, construct an appropriate energy-level diagram to calculate ΔH_r.

	$\Delta H^{\ominus}$/ kJ mol^{-1}
First ionisation energy of H(g)	+1310
First electron affinity of C*l*(g)	– 364
Standard enthalpy change of hydration of Cl^- (g)	– 381
Standard enthalpy change of hydration of H^+(g)	– 1130

[4]



By Hess' Law,

ΔH_r = ΔH_f (HC*l*(g)) + BE(H-C*l*) + 1st I.E.(H) + 1st E.A.(C*l*) + ΔH_{hyd} (H^+(g)) + ΔH_{hyd} (Cl^-(g))

ΔH_r = (-92) + 431 + (+1310) + (– 364) + (– 1130) + (– 381)

BE(H-C*l*) = **-226 kJ mol^{-1}**

Elements should be clearly placed at ground state of energy = 0.

(c) When heated, Group 17 hydrides, HX, undergo thermal decomposition.

$$HX \longrightarrow \tfrac{1}{2} H_2 + \tfrac{1}{2} X_2$$

Using the data on bond energies of the various H-X bonds given below, predict and explain how the thermal stability of the hydrides varies down the Group 17.

Bonds	Bond energies/ kJ mol^{-1}
H-C*l*	+431
H-Br	+366
H-I	+299

[2]

The thermal stability of Group 17 hydrides **decreases** down the group.

Down the group, the **atomic radii of the halogen increases**, which resulted in the less effective overlap due to the bigger orbitals involved. The **H-X bond length increases,** resulting in decreasing bond energy and lesser energy is required to break the weaker H-X bond for decomposition and the thermal stability decreases.

(d) Seawater is an important source of ionic halides like sodium chloride, sodium bromide and sodium iodide.

When silver nitrate is added to separate test-tubes containing aqueous sodium chloride, sodium bromide and sodium iodide, the various AgX (X = C*l*, Br and I) precipitates were observed. The solubilities of these precipitates upon addition of dilute NH_3 and concentrated aqueous NH_3, are recorded below.

	NaC*l*	**NaBr**	**NaI**
To NaX(aq), add $AgNO_3$(aq)	white ppt observed	cream ppt observed	yellow ppt observed
To the resulting mixture, add dilute NH_3(aq)	soluble	insoluble	Insoluble
To the resulting mixture, add excess concentrated NH_3(aq)	soluble	soluble	Insoluble

(i) Write the balanced ionic equation for the formation of the cream precipitate from adding silver nitrate to sodium bromide.

[1]

$Br^-(aq) + Ag^+(aq) \rightarrow AgBr(s)$
State symbols must be given as it is ionic equation.

(ii) Explain the effect of adding aqueous ammonia to each of the resulting mixture and clearly account for any difference in the observation. [3]

$AgX(s) \rightleftharpoons Ag^+(aq) + X^-(aq)$
$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$ **reaction 2**

The **solubility product, K_{sp}, of the silver halides decreases down Group 17** (i.e. K_{sp} of AgC*l* > AgBr > AgI).

When aqueous ammonia solution is added, based on reaction **(2)**, the small amount of Ag^+(aq) ions present in the resulting mixture is used up to form $[Ag(NH_3)_2]^+$(aq) complex. This **decreases the concentration of Ag^+(aq) ions** as a result and **the ionic product of the silver halide decreases** .

Since silver chloride has the largest K_{sp}, the **ionic product** drops below **K_{sp} of AgCl** and hence silver chloride dissolves when NH_3(aq) solution is added.

In contrast, silver bromide and sodium iodide has smaller K_{sp}. Hence, even though ionic product decreases, the **ionic products of both silver bromide and silver iodide are still larger than their respective K_{sp} values** and the precipitates remained insoluble when dilute NH_3(aq) solution is added.

However, with concentrated aqueous ammonia, the formation of diamminesilver(I) complex is sufficient to decrease the [Ag^+], thus causing the ionic product of AgBr to fall below its K_{sp} value thus ppt dissolves. This explains why AgBr(s) is soluble in concentrated NH_3(aq) but insoluble in dilute NH_3(aq).

For AgI:
The formation of diamminesilver(I) complex in both dilute and concentrated aqueous ammonia is not sufficient to cause the ionic product of AgI to fall below its K_{sp} value (as the K_{sp} value of AgI is extremely low). Therefore, AgI(s) is insoluble in both dilute and concentrated NH_3(aq).

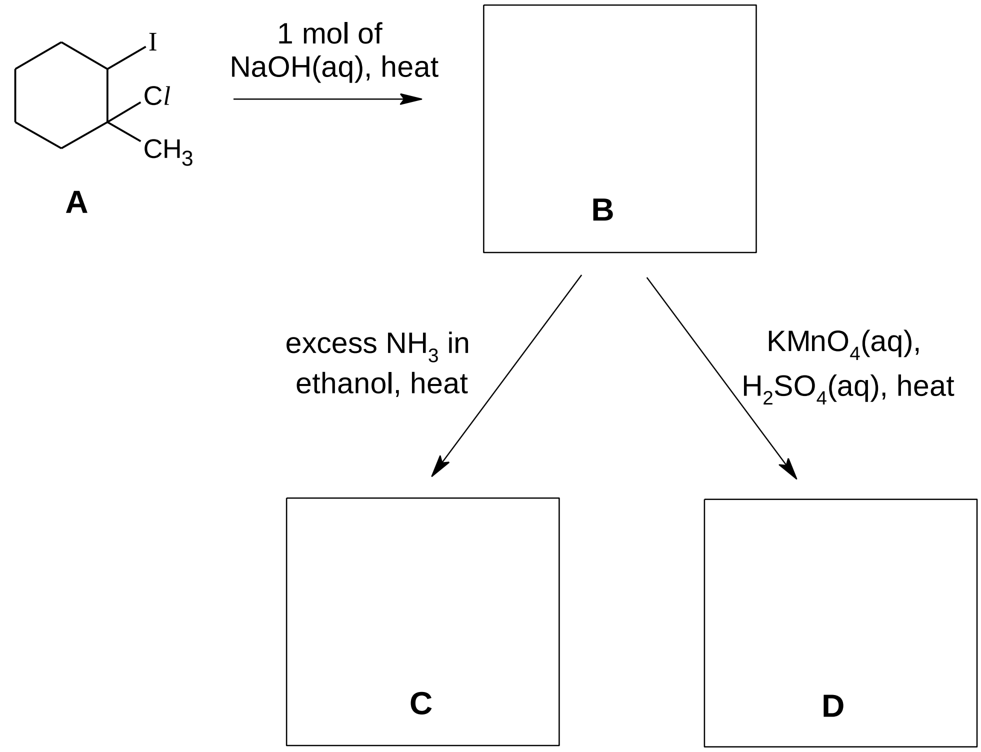
(e) Halogenoalkanes are widely synthesised by reacting halogens with alkenes. In turn, halogenoalkanes can undergo nucleophilic substitution using sodium hydroxide to form alcohols commercially.

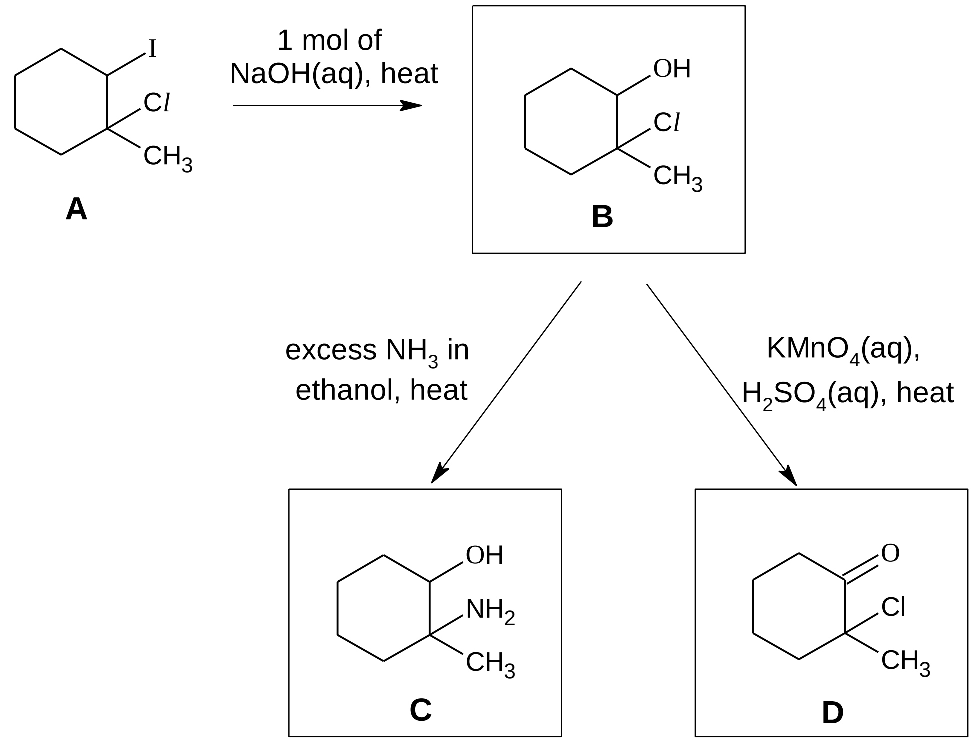
(i) Give one reason why iodoalkanes are more reactive towards nucleophilic substitution with aqueous NaOH than the corresponding chloroalkanes.

[1]

The bond energy of C–I is lower / C–I bond weaker than C– Cl. C–I bond is more easily broken.
OR
Cl^- is also a weaker leaving group than I^- as Cl^- is a stronger base than I^-.

(ii) A sequence of reactions, starting from compound **A**, a dihalogeno compound, is shown below.



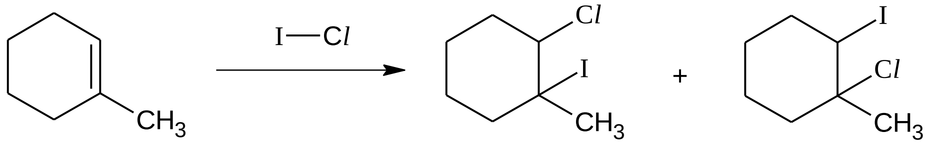


Draw the structures of compounds **B**, **C** and **D**.

[3]

(iii) The halogens form many interhalogen compounds, one such example is ICl, commonly known as Wijs' reagent.

Wijs' reagent can react with 1-methylcyclohexene to form a mixture of compound **E** and **A**. Explain why **A** is the major product for this reaction.



1-methylcyclohexene **E** **A**

[2]

The alkene functional group of 1-methylcyclohexene will **attack the iodine atom** of the I*Cl* molecule **first as iodine atoms are more delta positive** than C*l*, to form a new C- I bond. The major product will have the new IC*l* bond formed on the carbon number 2, as this will result in a **more stable carbocation with the positive charge on the first carbon** which had an additional **electron-donating methyl group**. OR form more stable carbocation

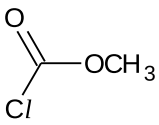
(f) **(i)** Below outlines a reaction scheme of phosgene with NaOH(aq) in two stages.

$$COCl_2 \xrightarrow[\text{stage 1}]{OH^-} COCl(OH) \xrightarrow[\text{stage 2}]{} CO_2 + HCl$$

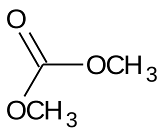
State the types of reaction occurring in stage 1, and stage 2. [2]

Stage 1: Nucleophilic acyl substitution (*for carboxylic acids and derivatives*)
Stage 2: Elimination

(ii) In order to remove any phosgene present in a bottle of chloroform, a small amount of methanol is added. use the information in **(f)(i)** to suggest the structural formula of the organic compound formed when phosgene reacts with methanol. [1]



or



[Total:20]

2 **(a)** Carbon monoxide reacts with steam over a copper / zinc catalyst, to form a mixture of carbon dioxide and hydrogen gas.

Reaction 1: $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

When an equimolar mixture of hydrogen and carbon dioxide at an initial total pressure of 4 atm is allowed to reach equilibrium at 1000 K, the percentage of steam in the mixture of gases is found to be 15%.

(i) Calculate K_p of the system in **reaction 1** at 1000 K. [2]

	$CO(g)$	$H_2O(g)$	$\rightleftharpoons$	$CO_2(g)$	+	$H_2(g)$
Initial pressure / atm	0	0		2		2
Change in pressure / atm	+x	+x		-x		-x
Eqm pressure / atm	x	x		2-x		2-x

$P_{H2O} = x / 4 = 0.15$
Thus, $x = P_{H2O} = P_{CO} = 0.60$ atm
$P_{CO2} = P_{H2} = 1.40$ atm
$K_p = (1.40)^2 / (0.60)^2 = 5.44$

(ii) Suggest, with a reason, whether or not a higher pressure would favour the formation of steam. [1]
No because same number of moles of gas on right and left side

(b) Sodium ethanoate can be used to prepare ethane by an electrochemical reaction, which is known as the Kolbe electrolysis reaction. Ethane is formed at the anode by the following reaction:

$$2CH_3CO_2^- \rightarrow C_2H_6 + 2CO_2 + 2e^-$$

A student investigated the electrolysis of aqueous solution of sodium ethanoate, using graphite electrodes in a litmus solution.

(i) Give the equation for the reaction occurring at the cathode, and state the final colour expected at the cathode.

Construct an equation for the **overall** reaction. [2]

At the cathode:
- Equation at cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$
- The product OH^- turns the litmus solution blue.

- Overall reaction:
$$2\,CH_3CO_2^- + 2H_2O \rightarrow C_2H_6 + 2CO_2 + H_2 + 2OH^-$$

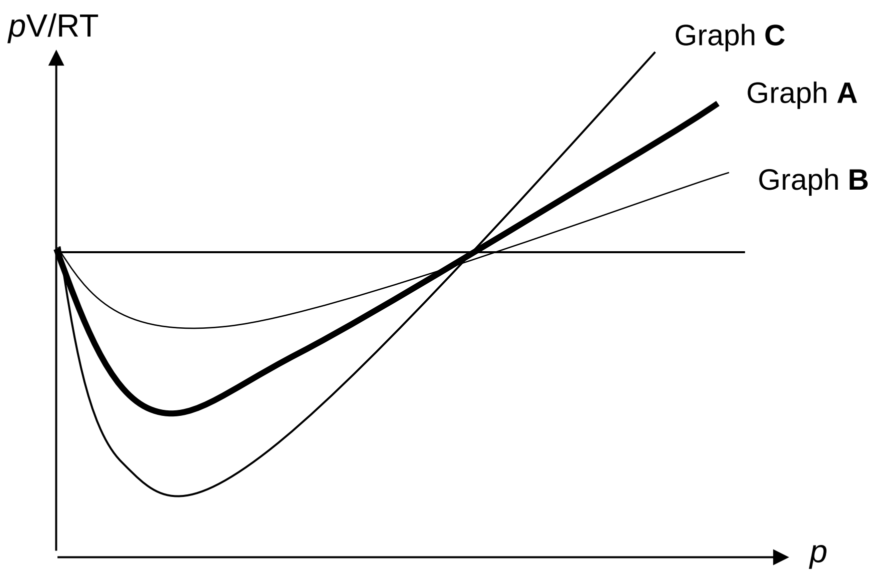
(ii) Calculate the time, in minutes, needed by the student to pass a current of 5.0 A through a solution of sodium ethanoate, to produce 1.50 g of ethane. [2]

No of moles of ethane = 1.5/30 = 0.0500 mol
No of moles of electrons = 0.05 x 2 = 0.100 mol

time = Q/I = $\frac{0.1\times96500}{5\times60}$ = 32.2 min

(c) **(i)** The value of pV/RT is plotted against p for the following three gases, where p is the pressure and V is the volume of the gas.
Given that graph **A** represents 1 mol of CO_2 at 298 K.

Identify which of the following graphs, **B** or **C,** represent 1 mol of CO_2 at 500 K.



[2]

Graph **B** is <u>CO_2 at 500 K</u>
CO_2 at 500 K deviates less than CO_2 at 298 K. At higher temperature, CO_2 <u>molecules possess higher average kinetic energy</u> and are more able to <u>overcome forces of attraction</u> between the molecules.

(ii) Sulfur dioxide is used by the Romans in winemaking, when they discovered that burning sulfur candles inside empty wine vessels keeps them fresh and free from vinegar smell.

One such cylinder of sulfur dioxide has an internal volume of 2.5 dm^3, and a mass of 2.3 kg.

Calculate the pressure (in pascals) that the sulfur dioxide would exert inside the cylinder at room temperature. [1]

Pls take note: Room temperature is now 293K!
pV = nRT
p = $(2.3 \times 10^3 / 64.1)(8.31)(293) / (2.5 \times 10^{-3})$
= 3.49×10^4 kPa

(d) Sulfuric acid is used in many industrial processes of major importance.

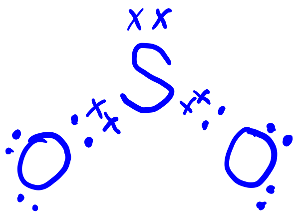
The process used to produce sulfuric acid involves a three–stage process. The first stage is to pass air over burning sulfur. The emerging gas is then passed over a catalyst, V_2O_5, which is maintained at 450-550ºC in the reaction chamber.

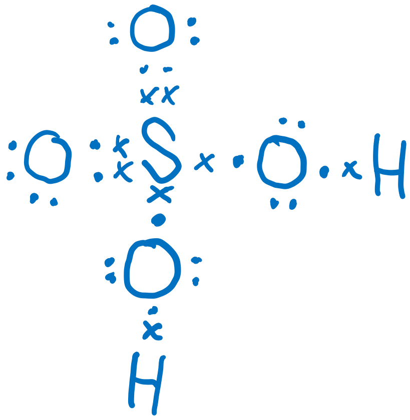
Stage **I**: $S(s) + O_2(g) \rightarrow SO_2(g)$

Stage **II**: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Stage **III**: $SO_3(g) + H_2O(l) \rightarrow H_2SO_4(l)$

(i) Draw dot-and-cross diagrams to show the bonding in SO_2 and H_2SO_4. State the bond angle, with respect to sulfur atom, in each of them. [4]





Accept between 110º and 120º

109.5º

(ii) The boiling points of sulfur trioxide and sulfur dioxide are 45ºC and -10ºC respectively.

Account for the difference in boiling points for both compounds, in terms of structure and bonding. [2]

- Both compounds have **simple covalent structures/ simple molecular structures**.
- However, SO_3 has a higher boiling point than SO_2 because SO_3 has **stronger (intermolecular) instantaneous dipole-induced dipole attractions** due to its
- **larger electron cloud size / more electrons being polarised.**

(SO_2 has weaker intermolecular permanent dipole-dipole attractions.)

- Hence, **more energy** is required to overcome the bonding in SO_3.

(e) For each of the following pairs of compounds, suggest **one** simple chemical test to distinguish them from each other. State the reagents and conditions needed and give the expected observations for each compound.

(i)

$CH_2{=}CHCH_2CH(CH_3)OH$	$CH_2{=}CHC(CH_3)_2OH$
J	**K**

Test: Add **$K_2Cr_2O_7$(aq), H_2SO_4(aq) and heat** to both compounds**.**

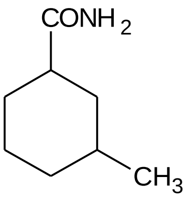
Observation: The solution of **A** will change from **orange to green** while the solution of **B** will **remain orange**.

OR

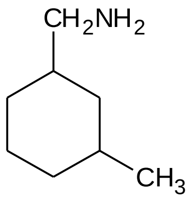
Test: Add **I_2(aq), NaOH(aq) and warm** to both compounds.

Observation: The solution of **J** will decolourise brown iodine and give a **yellow ppt** while solution of **K** **will not**.

(ii)



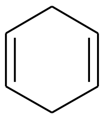
L



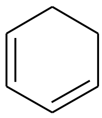
M

Test: Add NaOH(aq) and heat to both compounds.
Observation: **L** will give pungent gas (NH_3) that turns **moist red paper blue** while **M will not**.

(iii)



P



Q

Add $KMnO_4$(aq), H_2SO_4(aq) and heat to both compounds**.**
Observation: The solution of **P and Q** will change from **purple to colourless** but the solution of **Q** will **produce** bubbles of CO_2 that form white ppt with limewater

[Total:22]

3 Nitrous oxide, N_2O, is commonly known as laughing gas due to the euphoric effects of inhaling it. At a temperature of 1000 K, in the presence of chlorine catalyst, N_2O decomposes to its elements according to the following equation:

$$2N_2O(g) \rightarrow 2N_2(g) + O_2(g)$$

The experimental rate equation for this reaction is Rate = $kP^{a}_{N_2O}\,P^{b}_{Cl_2}$

where P_{N_2O} and P_{Cl_2} are the partial pressures of N_2O and Cl_2 respectively and a and b are non-zero constants.

The rate of decomposition of pure N_2O may be followed at constant temperature, by measuring the total pressure of the system with time.

The data below refers to a reaction in which pure N_2O decomposes in the presence of chlorine catalyst, Cl_2, (at partial pressure of 47 kPa).

Experiment 1:

time / s	0	40	80	140	240
P_{total} / kPa	87.0	91.3	93.0	98.0	102.2
P_{N_2O}	40	31. 4	25	18	9.6

(a) **(i)** Show that the partial pressure of N_2O at any instant is given by
$P_{N_2O} = 214 - 2P_{total}$ [2]

Let x = pressure of N_2O that dissociated

	$2N_2O(g)$	$\rightarrow 2N_2(g)$	$+ O_2(g)$

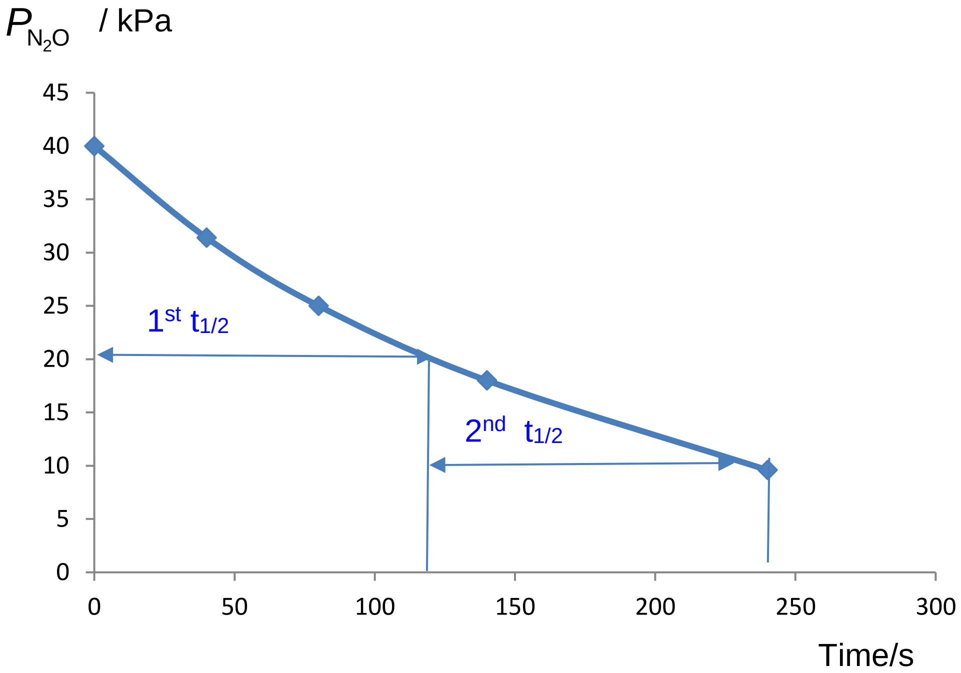
Initial P / kPa	40	0	0
Change in P/ kPa	-2x	+2x	+1x
Final P / kPa	40-2x	2x	X

P_{total} = 40-2x + 2x + x +47 = 87+x

X = P_{total} - 87

P_{N_2O} = 40-2x = 40 -2(P_{total} – 87) = 214 -2P_{tot} (shown)

(ii) Use the given data provided in the table above to plot a graph of P_{N_2O} against time on suitable axes. Showing all your working and drawing clearly any construction lines on your graph, use your graph to determine the value of a, the order of reaction with respect to N_2O.



-Axes, correct labelled (with units)
-Correct plotting of points
-Correctly draw grid lines to show the two $t_{1/2}$ (range of half-life ±10%)
$t_{1/2}$ is about 121s
-Conclude first order from constant half lives construction lines for the 2 half-lives [4]

(iii) Two additional experiments, experiments 2 and 3 were carried out and the following results were obtained.

	P_{N_2O} / kPa	P_{Cl_2} / kPa	Initial rate/ kPa s^{-1}
experiment 2	70	35	0.306
experiment 3	70	70	0.613

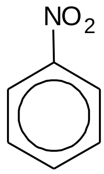
Determine b, the order of reaction with respect to chlorine. Explain your reasoning. [1]

Compare experiments 2 and 3, keeping P_{N_2O} constant, when P_{Cl_2} doubled, initial rate doubled. Order with respect to chlorine =1

(iv) Hence write the rate equation for the reaction. [1]

Rate = $k P_{N_2O}\, P_{Cl_2}$ give ecf based on students' orders from (e)(ii) and (iii)

(b) Nitrobenzene is an organic compound used to mask unpleasant odors in shoe and leather dressings.



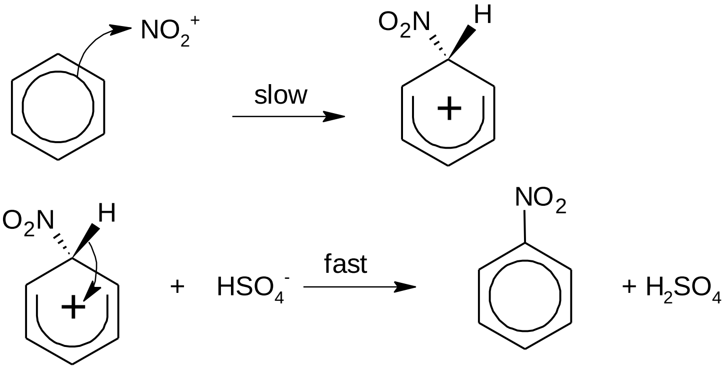
(i) Suggest the type of hybridisation of N in nitrobenzene. [1]

sp^2 hybridisation

(ii) Nitrobenzene can be formed from reacting benzene with concentrated nitric acid and sulfuric acid. Name and outline the mechanism of the nitration of benzene. Indicate the rate-determining step and draw the formula of the organic intermediate. [3]

Electrophilic substitution

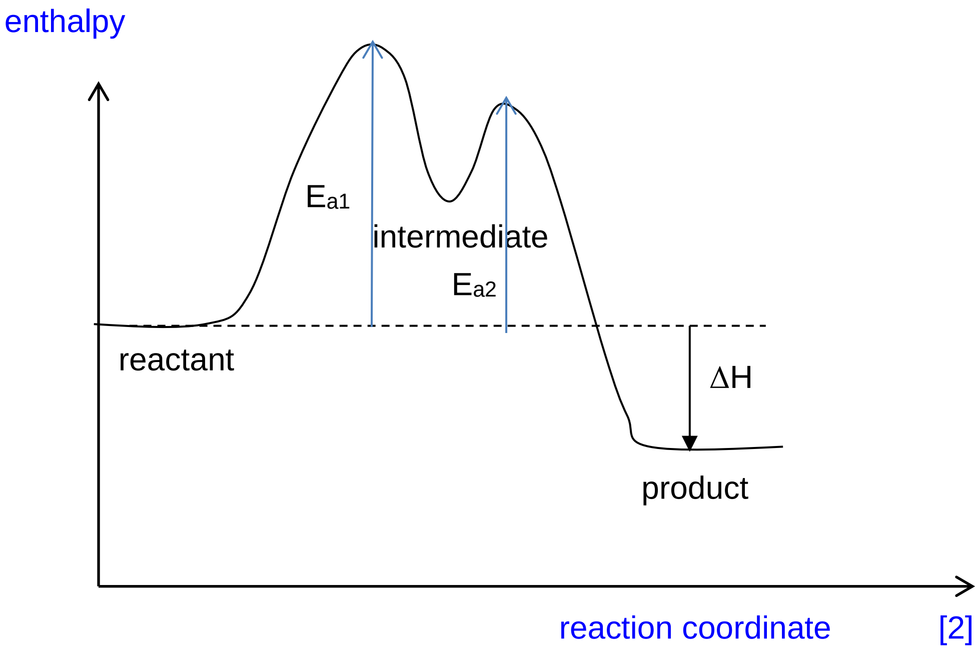
$2H_2SO_4 + HNO_3 \rightleftharpoons 2HSO_4^- + NO_2^+ + H_3O^+$



Deduct one mark if any of these are missing below

- Correctly drawn curly arrows for step 1
- Correctly drawn curly arrows for step 2
- Label **slow (rate-determining)** step
- correct intermediate (no need to show dashed and bold lines)
- Regeneration of catalyst**, H_2SO_4**

(iii) Given that the nitration of benzene is exothermic, sketch the reaction pathway diagram of the reaction, labelling clearly the reactant, intermediate and product on your diagram.



[2]

- Axes clearly labelled
- Relative energy level of reactants and product (lower)
- Show 2 ‘humps’, and Ea1 > Ea2
- Label reactant,
- Label intermediate and
- Label product

(c) To investigate the mechanism of nitration of benzene, every hydrogen atom on benzene is replaced by its isotope, deuterium, to form C_6D_6.

Carbon–deuterium (C-D) bond has higher bond energy than a carbon–hydrogen (C-H) bond.

If the rate-determining step involves the breaking of a C–H bond, replacing the C–H bond with a C–D bond will change the rate constant, *k*, of the reaction. This is known as the deuterium isotope effect.

Deuterium isotope effect = $\frac{k_H}{k_D}$

k_H = rate constant for nitration of C_6H_6

k_D = rate constant for nitration of C_6D_6

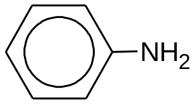
If $\frac{k_H}{k_D}$ = 1, it is said that the deuterium isotope effect is absent.

If $\frac{k_H}{k_D}$ >> 1, it is said that the deuterium isotope effect is present.

Predict with reasoning, if the deuterium isotope effect would be present in the nitration of benzene. [1]

As the rate-determining step does not involve the breaking of a C-H bond / the C-H bond is only broken in the fast step of the mechanism, there is no deuterium isotope effect present in nitration of benzene.

(d) **(i)** Nitrobenzene can be converted to phenylamine, which is used in the synthesis of polyurethane, a polymer used in making tyres.



phenylamine

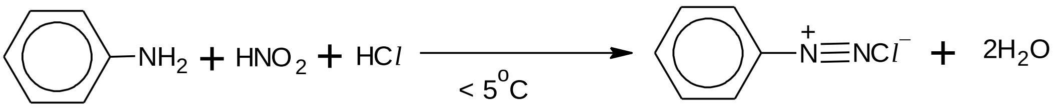
Suggest the reagents and conditions to convert nitrobenzene to phenylamine. [1]

Sn, concentrated HCl, heat
Followed by NaOH(aq)

(ii) The following equations illustrate the formation of phenol from phenylamine, in two steps:

In step 1, phenylamine reacts with cold nitrous acid, HNO_2, and hydrochloric acid, HC*l*, to form phenyldiazonium chloride.

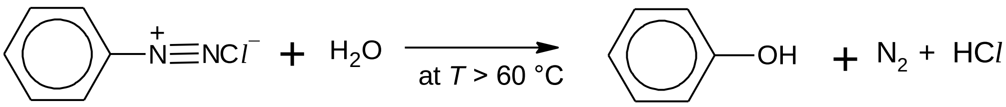
Step **1**



phenyldiazonium chloride

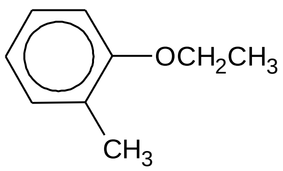
In step 2, phenyldiazonium chloride can react with water upon heating to give phenol.

Step **2**



phenyldiazonium chloride

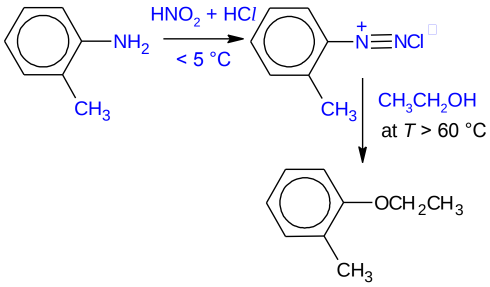
Propose a two-step synthetic pathway for the conversion of 2-methylphenylamine to compound **S** below.



Compound **S**

[2]

Answers:



[Total:18]

Section **B**
Answer **one** question from this section.

4 (a) A blood test can detect abnormally high levels of an amino acid which is one of the signs of the disease phenylketonuria, a rare condition in which a person is unable to properly break down amino acids such as phenylalanine.

In acidic solution, phenylalanine is completely protonated and exists as $H_3N^+CHRCO_2H$, where R is $–CH_2C_6H_5$.

Protonated phenylalanine, $H_3N^+CHRCO_2H$, acts as a dibasic acid that ionises in stages.

$H_3N^+CHRCO_2H \rightleftharpoons H_3N^+CHRCO_2^- + H^+$ $pK_1 = 1.83$

$H_3N^+CHRCO_2^- \rightleftharpoons H_2NCHRCO_2^- + H^+$ $pK_2 = 9.13$

(i) Calculate the pH of a 0.80 mol dm^{-3} solution of protonated phenlyalanine (ignore the effect of pK_2 on the pH).

Let $[H_3O^+] = [H_3N^+CHRCO_2^-] = x$ mol dm^{-3} at equilibrium

	$H_3N^+CHRCO_2H$	+	H_2O	$\rightleftharpoons$	H_3O^+	+ $H_3N^+CHRCO_2^-$
Initial conc	0.80				-	-
Change in conc	-x				+x	+x
Eqm conc	0.80-x				x	x

$$K_a = \frac{[H_3O^+][H_3N^+CHRCO_2^-]}{[H_3N^+CHRCO_2H]} = \frac{x^2}{0.80 - x}$$

Since protonated phenylalanine is a weak acid, extent of dissociation is small.
⇒ Hence x is negligible compared to 0.80 mol dm^{-3}.
Therefore, $(0.80 – x) \approx 0.80$

$$K_a = \frac{x^2}{0.80} = 10^{-1.83}$$

Solving, x =0.10878 mol dm^{-3}

Hence, initial pH = $-\lg [H^+ (aq)]$ = 0.963 (3sf)

(ii) An amphiprotic species is one that can act as a Bronsted-Lowry acid or base. The pH of a solution containing an amphiprotic species is given by the following expression.

$pH = ½(pK_1 + pK_2)$

In the titration of protonated phenylalanine with NaOH, an amphiprotic species is formed.

Identify the amphiprotic species formed and calculate the pH of the solution.

Amphiprotic species formed = $H_3N^+CHRCO_2^-$

pH at 1st equivalence point= (1.83 + 9.13)/2 = 5.48

(iii) Sketch the titration curve when 25.0 cm^3 of protonated form of phenylalanine is being titrated with 70 cm^3 of NaOH(aq) of the same concentration. On your sketch, clearly mark the two pK_a values and the points you have calculated in **(i)** and **(ii)**.

- Label both axes
- Label initial pH = 0.963
- Sketch shape correctly 2 sharp changes in pH at correct equivalence volumes 25.0 cm^3 & 50.0 cm^3
- Label pK_1 = 1.83 at 12.5 cm^3
- Label pK_2 = 9.13 at 27.5 cm^3
- pH at 1st equivalence point = 5.48

(iv) Suggest a suitable indicator from the following table to be used to detect the first equivalence point and state the colour change of the solution at this equivalence point.

indicator	pH range	acid solution	basic solution
Thymol Blue	1 – 3	Red	yellow
Methyl Red	5 – 6	yellow	Red
Phenol red	7 – 8	yellow	Red
Phenolphthalein	8 – 10	colourless	red

1st equivalence point pH = 5.48 (region of sharp pH change), methyl red is a suitable indicator.
The colour will change **from yellow to orange**.

(b) "Acidity regulators" are food additives that have a buffering action on the pH of food. A mixture of citric acid, $C_5H_7O_4CO_2H$, and its sodium salt is often used for this purpose.

You may assume that citric acid behaves as a monobasic weak acid.

$C_5H_7O_4CO_2H \rightleftharpoons C_5H_7O_4CO_2^- + H^+$; $K_a = 7.4 \times 10^{-4}$ mol dm^{-3}

(i) A citric acid / sodium citrate buffer mixture is prepared by mixing 40.0 cm^3 of 0.100 mol dm^{-3} citric acid and 60.0 cm^3 of 0.200 mol dm^{-3} sodium citrate.

Calculate the pH of the buffer solution. [1]

Amount of HA = 4 x10^{-3} mol
Amount of A^- 0.012 mol

pH = pKa + log [A^-]/[HA]

= 3.13 + log (3) = 3.63

(ii) Write equations to show how this mixture of citric acid and sodium citrate regulates the acidity on addition of H^+ ions and OH^- ions. [2]

On addition of H^+,

$C_5H_7O_4CO_2^- + H^+ \longrightarrow C_5H_7O_4CO_2H$

On addition of OH^-,

$C_5H_7O_4CO_2H + OH^- \longrightarrow C_5H_7O_4CO_2^- + H_2O$

Full forward arrows

(c) **(i)** Describe the conditions needed to hydrolyse a protein non-enzymatically. [1]

Heat under reflux with H_2SO_4(aq) or NaOH(aq) for several hours.

(ii) Histidine decarboxylase undergoes enzymatic hydrolysis. Upon hydrolysis, the polypeptide fragments of the enzyme's active site are isolated as shown below.

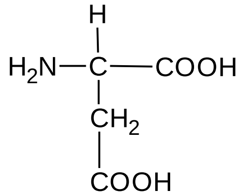
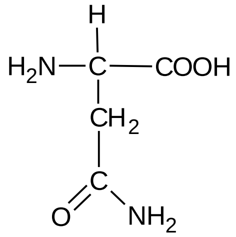
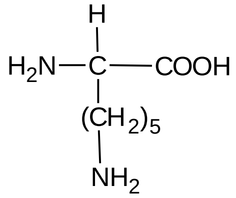
- Ala-Cys-Phe
- Gly-Gly
- Lys-Asp-Asp-Gly-Gly
- Phe-Arg-Lys

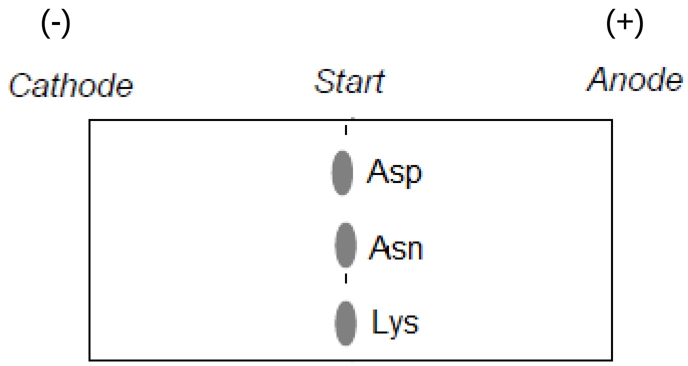
Deduce the sequence of the 9 amino acid residues of the enzyme's active site. [2]

Working
Ala-Cys-Phe
Phe-Arg-Lys
Lys-Asp-Asp-Gly-Gly
Gly-Gly must show working for stacking

Ala-Cys-Phe-Arg-Lys-Asp-Asp-Gly-Gly

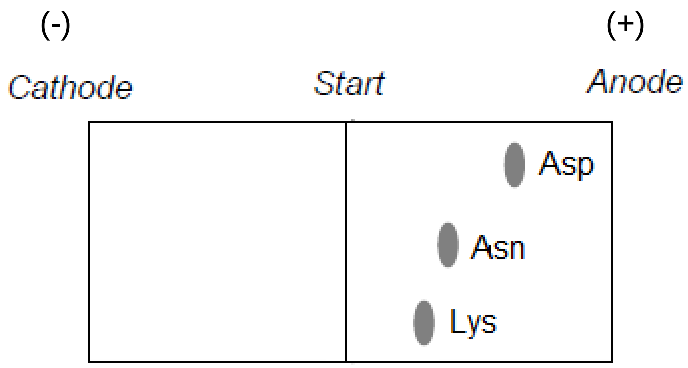
(iii) A different segment of tripeptide, Asp-Asn-Lys, was subjected briefly to acidic hydrolysis which produced individual amino acids as well as various peptides. The resulting mixture buffered at pH 13 was separated in an electric field using electrophoresis.

Amino acid			
Abbreviation	Asp	Asn	Lys



Using the above template sketch and label the position of the three amino acids after the separation. State one factor that affect the rate at which the charged particles move during electrophoresis.

[3]

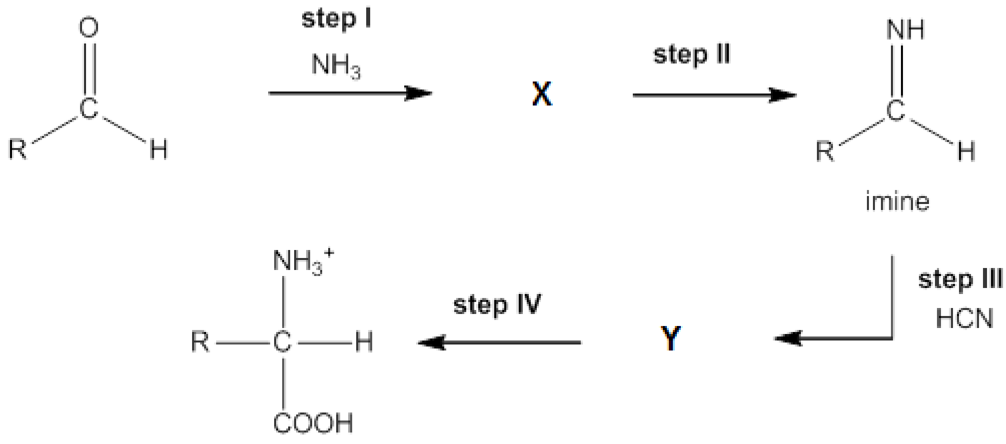


One factor: Mass or charge density of species

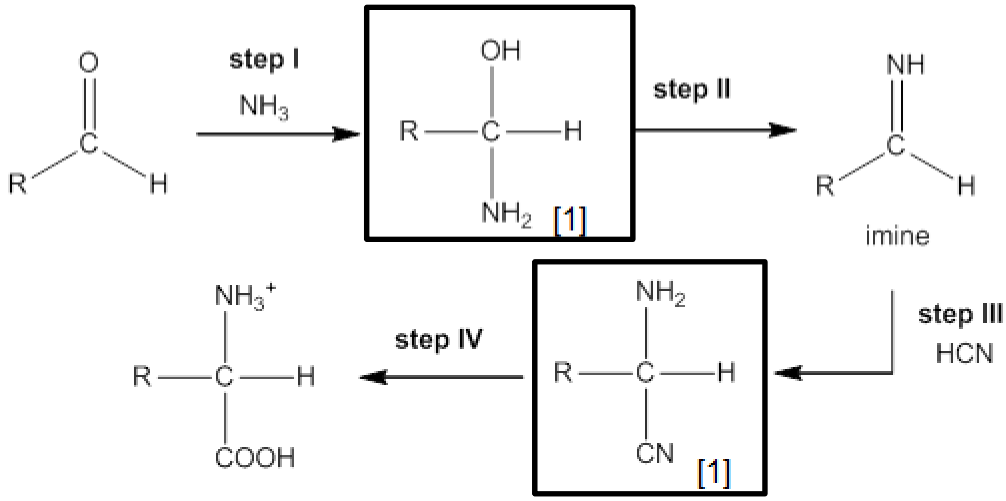
(d) The Strecker synthesis assembles an α-amino acid from the amine precursor and an aldehyde.

- Step **I** is a nucleophilic addition reaction.
- Step **II** involves the elimination of water from **X** to form an imine**.**
- Step **III** is similar to the formation of cyanohydrin when the imine undergoes nucleophilic addition with HCN and forms an α-amino nitrile, **Y**.
- Step **IV** is the hydrolysis of **Y** to form the carboxylic acid functional group.

In the following reaction flow scheme, draw the intermediate compounds **X** and **Y** and suggest the reagent and condition for the step **IV** in the scheme.



[3]



Step IV: H_2SO_4 (aq) heat or HC*l* (aq) heat (accept dilute in place of aq)

[Total:20]

5 Copper is an important transition element that can readily form complex ions with *ligands*. Copper forms a large variety of compounds, usually with variable oxidation states. Copper compounds also act as useful *homogeneous catalysts*, commonly used in many industrial operations.

(a) Define the terms *ligand* and *homogeneous catalyst*. [2]

Ligand is a neutral molecule or anion which contains at least **1 lone pair of electrons** available for forming **dative bonds** with the available **empty orbitals** on the central metal ion/atom.

Homogeneous catalysts **increase the rate of reaction** by providing an alternative pathway with **lower activation energy**. *Homogeneous catalysts* are in the **same phase as the reactants** in the reaction.

(b) In recent years, horseshoe crabs have generated great interest amongst scientists due to its blood with remarkable antibacterial properties. An unusual aspect of the blood is that it is bright blue, a consequence of using copper-based haemocyanin to transport oxygen.

Haemocyanin (Hc) contains two copper atoms that reversibly bind a single oxygen molecule, O_2, to form oxyhaemocyanin, HcO_2, according to the following equation:

$$2Hc(aq) + O_2(aq) \rightleftharpoons 2HcO_2(aq) \quad \textit{equilibrium 1}$$

Oxygenation causes a color change between Cu(I) deoxygenated form and the Cu(II) oxygenated form.

(i) By considering *equilibrium 1*, comment on how the crab cells receive oxygen from the air in the lungs. Check TYS haemoglobin
Hint: Consider the level of oxygen in the lungs

[2]

In the lungs, $[O_2]$ is high. This results in the **position of equilibrium shifting to the right**, favouring the formation of the oxyhaemocyanin HcO_2.

In the cells, $[O_2]$ is low. The position of equilibrium will hence **shift to the left**, favouring the release of O_2.

(ii) Indicate the color change from Cu(II) oxygenated form to Cu(I) deoxygenated form. Hence explain why both copper complex ions exhibit different colours.

[5]

From blue to colourless

oxyhaemocyanin contains Cu^{2+}: [Ar] $3d^9$
Cu(II) has partially filled 3d-orbitals.
In presence of ligands, the originally degenerate 3d orbitals split into 2 sets of different energy levels, separated by a small energy gap (ΔE).

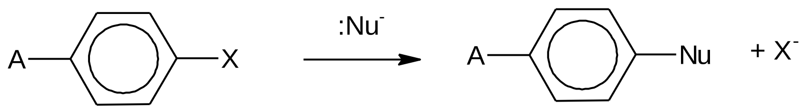
ΔE corresponds to the energy of visible light. An electron from the lower energy d orbital absorbs a specific wavelength of visible light and is promoted to a half-filled, higher energy 3d orbital. d-to-d transition can take place and colour observed is complementary to the colour absorbed.

deoxygenated haemocyanin contains Cu^+: [Ar] $3d^{10}$
Cu(I) has fully filled 3 d-orbitals ($3d^{10}$). Hence, d-to-d electron transition cannot occur, and no visible light is absorbed, which accounts for it being colourless.

(c) Most aromatic compounds undergo electrophilic substitution.

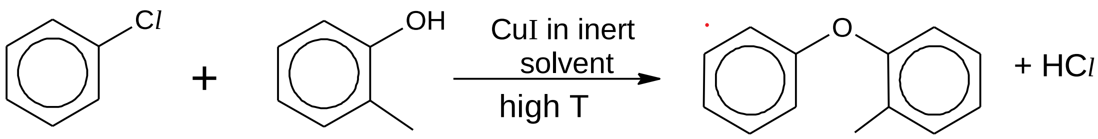
However it is possible for aryl halides to undergo a limited number of nucleophilic substitution reactions with strong nucleophiles.

An example of a reaction is as follows:



An example of an aryl nucleophilic substitution is the *Ullmann coupling* reaction where hydrocarbons fragments are coupled, using copper compounds as a catalyst.

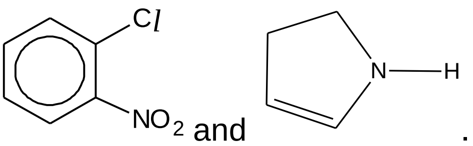
An example is indicated in reaction **1** below.



reaction **1**

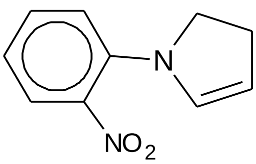
(i) The conditions for the *Ullmann coupling* reactions are usually harsh conditions, and yet give low yield.

In a similar reaction under the same temperature and pressure, the two compounds below were reacted together:



Deduce and draw the structure of the product for the above reaction.

[1]



(ii) Predict, with reasoning, if the above reaction in **(c)(i)** will result in a higher or lower yield compared to reaction **1**.

[2]

The presence of the electron-withdrawing NO_2 group on the benzene increase the $\delta+$ on the carbon which attracts nucleophile more strongly

As such, the above reaction will result in a higher yield.

(iii) Like copper(I) iodide, aqueous Fe^{2+} is a good homogeneous catalyst used in the decomposition of aqueous hydrogen peroxide as indicated below.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

With reference to relevant data in the *Data Booklet*, suggest a mechanism for the catalysis of the above reaction by aqueous Fe^{2+}. [3]

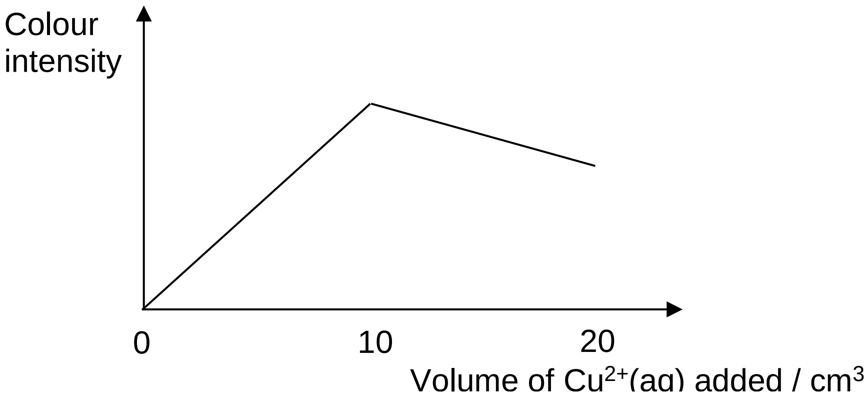
$H_2O_2 + 2H^+ + 2e \rightarrow 2H_2O$ $E^\theta = +1.77V$
$Fe^{2+} \rightarrow Fe^{3+} + e$ $E^\theta = -0.77V$
Step 1 Overall: $H_2O_2 + 2H^+ + 2Fe^2 \rightarrow 2H_2O + 2Fe^{3+}$ $E^\theta_{cell} = +1.00V$
mark for calculating E^θ_{cell}]

$H_2O_2 \rightarrow O_2 + 2H^+ + 2e$ $E^\theta = +0.68V$
$Fe^{3+} + e \rightarrow Fe^{2+}$ $E^\theta = +0.77V$
Step 2 Overall: $H_2O_2 + 2Fe^{3+} \rightarrow O_2 + 2Fe^{3+} 2H^+$ $E^\theta_{cell} = +0.09V$

(d) Copper(II) species can have various coordination numbers, though tetracoordinated copper(II) species are not as common as hexacoordinated ones.

The formula of a methylamine-copper complex ion, $[Cu(CH_3NH_2)_x(H_2O)_2]^{2+}$, can be determined using a colorimeter.

In an experiment, a solution of 0.050 mol dm^{-3} of Cu^{2+}(aq) was added 1 cm^3 at a time to 20 cm^3 of 0.100 mol dm^{-3} of aqueous CH_3NH_2. The colour intensity of the resultant solution after each addition was measured using a colorimeter. The following graph was obtained.

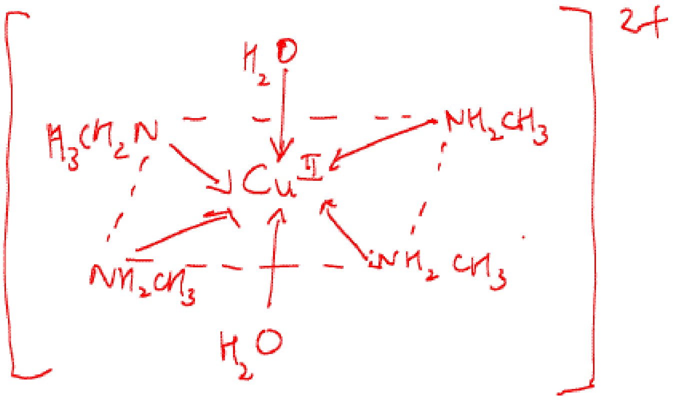


(i) Using the graph, determine the value of ***x***. [2]

No. of mol of Cu^{2+} = (0.10/1000) x 0.05 = 5×10^{-4}
No. of mol of CH_3NH_2 = (20/1000) x 0.1 = 2×10^{-3}

$1/x = (5 \times 10^{-4})/(2 \times 10^{-3})$
$x = 4$

(ii) Draw the structure of the methylamine-copper complex ion. [1]



(e) Mercury is poisonous as it can cause many health problems ranging from neurological, respiratory to cardiovascular problems. One possible treatment for mercury poisoning involves administering a solution of $EDTA^{4-}$, a common hexadentate ligand which forms complexes with many metal ions.

Copper and calcium ions are essential minerals required by the body. Using relevant data from below, comment on the use of $EDTA^{4-}$ as a treatment for mercury poisoning and comment on its effect on the concentrations of copper and calcium ions in the body.

Equilibrium	K_c /mol^{-1} dm^3
$Ca^{2+} + EDTA^{4-} \rightleftharpoons [Ca(EDTA)]^{2-}$	6×10^{10}
$Cu^{2+} + EDTA^{4-} \rightleftharpoons [Cu(EDTA)]^{2-}$	5×10^{18}
$Hg^{2+} + EDTA^{4-} \rightleftharpoons [Hg(EDTA)]^{2-}$	6.3×10^{21}

[2]

As K_c of $[Hg(edta)]^{2-}$ is higher than that of $[Ca(edta)]^{2-}$ and $[Cu(edta)]^{2-}$, Hg^{2+} will be **removed first** in preference of Cu^{2+} when edta is used. As such, it is useful in the removal of heavy metals (such as Hg) from the body.

However, when a high dosage of EDTA is added, Cu^{2+}, which are essential for health, will also be removed due to the shifting of the position of equilibrium to the right, favoring the formation of the complexes. (as K_c of $[Hg(edta)]^{2-}$ $[Cu(edta)]^{2-}$ are closer in magnitude)

Hence, limited amount of edta should be administered to prevent the loss of Ca^{2+} and Cu^{2+}.

[Total:20]

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